



wind in the generation mix in 2023 and expected to reach 32-33% in 2030), issues of intermittency and grid stability become more significant. To address these challenges, SECI has innovated/adopted Hybrid, Round-The-Clock (RTC), and Firm and Dispatchable RE (FDRE) models; and continues to do so. Let's break down these innovative concepts into simpler terms:

RTC implies a consistent supply of renewable energy round the clock, with availability as high as 70% on a monthly basis. Given the country's varied landscapes and weather conditions, this Category-I Miniratna is utilising RTC to harness RE more effectively by strategically mapping power generation to meet regional energy demands.

FDRE ensures that energy buyers receive the power they need when they need it, shifting the focus from simply using what's available to providing what's needed. This marks a significant shift in the relationship between energy buyers and suppliers.The guidelines are issued by the Ministry of Power/MNRE, and tenders are called accordingly, including FDRE. In these tenders, developers are tasked with designing RE plants to meet the specific demands of buyers, ensuring that at least 90% of the specified demand is met each month. Key features of FDRE tenders include power supply based on demand, with developers required to schedule power according to pre-fixed demand schedules provided by buyers. Additionally, developers must demonstrate a minimum "Demand Fulfilment Ratio" of 90% each month, with penalties for any shortfalls in meeting this requirement. The guidelines also allow for the sale of excess power to third parties or power exchanges, providing flexibility for developers.

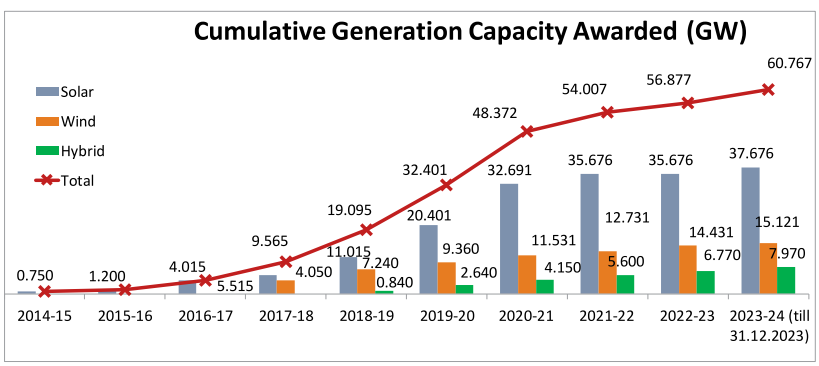
These initiatives represent SECI's concerted effort to anticipate and overcome the challenges associated with integrating renewable energy into India's electricity mix. This way, the company is unlocking the full potential of RE while ensuring stability and reliability in the power supply.

LOOKING AHEAD

SECI's unique role as a central power procurement intermediary has de-risked private investments, propelling the entire RE supply chain forward. By nurturing a favourable business environment, SECI has not only facilitated the implementation of RE projects but also fostered the growth of ancillary industries, thereby transforming the entire business ecosystem.

SECI is significantly contributing to the realisation of the ambitious agenda of 'Viksit Bharat @ 2047', articulated by Prime Minister Narendra Modi, envisioning a developed India by the 100th year of Independence. Central to this vision is the nation's Climate Agenda for 2047, a roadmap crafted to blend economic growth with environment-friendly industrial practices, resulting in a reduction of emission intensity and increase in clean energy generation.

As per NITI Aayog atleast, 1800 GW of RE capacity is expected to get installed by the year 2047. Of this, a significant portion is expected to come through SECI. The Corporation has a major role to play in some of the structural interventions required, such as: domestic manufacturing of RE technologies, promotion of energy storage, standardized bidding framework, de-risking RE investments and securing payments, that are crucial to meeting the country's 2047 targets.



With a focus on innovative models and a holistic approach, SECI is poised to lead the sector towards commercial viability. The transition from traditional solar and wind power to firm power delivery models, coupled with advancements in energy storage and transmission infrastructure, have further bolstered this CPSU's crucial role in the non-conventional energy sector.

As SECI continues to propel the nation's journey towards sustainable development, it is dedicated to meet the government's future RE targets, driving economic growth, and creating a sustainable future for generations to come.

> Against this backdrop, SECI's role becomes increasingly critical. As Rameshwar Prasad Gupta, Chairman and Managing Director, SECI succinctly puts it, "Aligned with Hon'ble PM Modiji's vision of 'Viksit Bharat @ 2047', SECI has emerged as a significant player in shaping India's RE landscape while adhering to the core principles of Aatmanirbhar Bharat and energy security. As the nation moves towards fulfilling the 2047 Climate Agenda, SECI remains steadfast in its commitment to steer the RE sector ahead. Through our innovative models and collaborative efforts with stakeholders, we are propelling the sector forward by fostering a stimulating business environment, competitiveness, and sustainability."

He adds, "We are poised to continue our role of helping the country meet its future RE targets while expanding into emerging areas such as green hydrogen and energy storage. Our ongoing and future initiatives promise to attract foreign investment and generate employment opportunities on a mega scale, further contributing to the realisation of a developed, self-reliant and energy secure India."